Fiscal Year:	FY 2021	**Task Last Updated:**	FY 09/04/2020
PI Name:	Sekulic, Dusan D.Sc.		
Project Title:	BRAzing of Aluminum Alloys IN Space		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Lexington	**State:**	KY
Zip Code:	40506	**Congressional District:**	6
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2015 NNH15ZTT002N MaterialsLab Open Science Campaigns for Experiments on the International Space Station
Start Date:	11/04/2016	**End Date:**	11/03/2021
No. of Post Docs:	2	**No. of PhD Degrees:**	2
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Strutzenberg Littles, Louise	**Contact Phone:**	(256) 544-0946
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 11/3/2021 per NSSC information (Ed., 3/30/21)		
Key Personnel Changes/Previous PI:	September 2020: No changes		
COI Name (Institution):	Krivilev, Mikhail Ph.D. (Udmurt State University, Russia) Mesarovic, Sinisa Ph.D. (Washington State University)		
Grant/Contract No.:	NNX17AB52G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	We propose to investigate the capillary flow phenomena relevant for techniques of brazing (bonding by molten alloy) in microgravity for space construction and repair of spacecrafts damaged by micrometeoroids and orbital debris. A series of space (International Space Station - ISS) and terrestrial experiments are planned to be integrated with modeling and analysis. These are proposed to answer the practical and fundamental questions: (1) How can we take advantage of the absence of gravity and dominance of capillary forces in transporting the large quantity of molten braze? (2) What composition of the brazing alloy and the brazing technique will enable strong and efficient bonding of structural elements and repair? (3) Can this brazing be done using flux-less cladding? The phenomena to be studied under these conditions include molten metal kinetics and wetting and spreading of liquid clad/filler over topographically modified surfaces. The applied aspects include direct verification if brazing of Al alloys is transferable to orbit for rapid repair of the pressure wall. For the first time, brazing in space will be processed using the patented flux-less Trillium technology. Further development includes design and testing of a space-based hot zone with the support from Marshall Space Flight Center (MSFC).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The impact of the proposed research on the life on Earth manifests itself through multifaceted benefits. These include: (i) uncovering so-far hidden physical/chemical/metallurgical aspects of metal bonding by brazing (extensively used in manufacturing) through uncoupling of the gravity impact on larger mass of molten aluminum driven by surface tension in gravity field, hence transforming the art of this technology to a predictive science for terrestrial applications, (ii) exploring a possibility to use novel materials as bonding alloys less sensitive on terrestrial hindrances (e.g., oxidation) in designing bonded assemblies and mitigating problems involving repair, (iii) featuring sustainable resources use and less energy invested due to an expected broadening the range of applicability of bonding/repair under adverse conditions, (iv) developing visualization of the brazing process in situ and in real time under controlled atmosphere, (v) developing the kinetics of wetting theory by implementing phase field approach and numerical techniques that can be used to model capillary phenomena not only for liquid metals but for other material systems in both micro gravity and gravity (on Earth) conditions.
Task Progress:	This document constitutes the Year 4 report of the research work conducted within the scope of the NASA sponsored Brazing Aluminum in Space (BRAINS) project (Grant NNX17AB52G). The work has included further studies of both experimental and theoretical modeling segments. The participation of both collaborating teams, the University of Kentucky (UK) and the Washington State University (WSU), established in Year 1, continues. A full participation of Udmurt State University (UdSU), Russia, continues as well. Both experimental and theoretical (modeling) works have been completed as planned at UK, WSU, and UdSU. New aspects of surface tension driven flow of molten aluminum phenomena have been considered. Modeling includes kinetics of molten alloy filling the hole configuration and flow in a wetting-nonwetting system with a smooth and rough substrates. The BRAINS team has completed the System Requirements Review (SRR), and been granted an advancement toward flight experiments. The sample design has been improved and finalized. Materials have been submitted for fabricating flight and ground sample ampoule assemblies for SUBSA (Solidification Using Baffle in Sealed Ampoules) tests. The modeling activities during the 4th year continued being focused on the following tasks/issues: (1) Modeling of surface tension-controlled equilibrium formations of molten aluminum, (2) Modeling of the kinetics of the process of filling the holes by surface micro layers of liquid aluminum alloy. Theoretical (modeling) work continues with further development of phase field theory which uncovers dynamic behavior of liquid metal flow during a hole-filling process. Results from the phase field model agree very well with empirical data.
Bibliography Type:	Description: (Last Updated: 12/06/2025)
Abstracts for Journals and Proceedings	Dehsara M, Fu H, Mesarovic SDj, Sekulic DP, Krivilyov M. "(In)Compressibility and parameter identification in phase field models for capillary flows." 2nd Biennial Society for Industrial and Applied Mathematics (SIAM) Pacific Northwest Section Conference, Seattle, WA, October 18-20, 2019. Abstracts. Society for Industrial and Applied Mathematics (SIAM) Pacific Northwest Section Conference, Seattle, WA, October 18-20, 2019. , Oct-2019
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Articles in Peer-reviewed Journals	Shutov IV, Kamaeva LV, Khamidullina AR, Krivilyov MD, Sekulic DP. "Phase transformations in thermal treatment of composite Al-Si+flux braze." Met Sci Heat Treat. 2020 Jan;62(7-8):498–501. https://doi.org/10.1016/j.jcrysgro.2019.125287 . [Article first appeared in Russian in the peer-reviewed journal Metallovedenie i Termicheskaya Obrabotka Metallov (MITOM). 2020 Jan;8:10-14.] , Jan-2020
Books/Book Chapters	Sekulic DP, Bassindale J. "Chapter P-9. Aluminum Controlled Atmosphere Brazing (CAB)." in "AWS Brazing Handbook, in print." Boca Raton, FL : American Welding Society, 2020. (in press as of September 2020), Nov-2020
Dissertations and Theses	Wu Y. [Wu, Yangyang] "Capillary Flow of Liquid Aluminum Alloy in Wetting and Wetting/Non-wetting Systems." PhD Dissertation, University of Kentucky, June 2020. https://doi.org/10.13023/etd.2020.273 , Jun-2020
Dissertations and Theses	Yu C-N. [Yu, Cheng-Nien] "Molten Aluminum Capillary Hole-Filling." PhD Dissertation, University of Kentucky, May 2020. https://doi.org/10.13023/etd.2020.168 , May-2020